Infosys - Coding 20

HackerRank

You are given a string S of length N over lowercase letters. You may delete any characters from S (each deletion costs D). After deletions, the string is run-length encoded: each maximal run of the same character c of length L is encoded as:

• cL — length $\lfloor \log_{10} L \rfloor + 2$, if L > 1

• c — length 1, if L = 1

The compression score is: N_original − encoded length − deletion cost. Find the maximum total compression score.

Input Format

First line: string s, the input string of length N (lowercase letters).

Second line: integer d, the cost per character deleted.

Constraints

1 <= |s| <= 100

1 <= d <= 20

Output Format

The output contains an integer value

Sample Input 0

```
aabbaa
1
```

Sample Output 0

```
2
```

Explanation 0

N=6, D=1. Keeping everything encodes as a2b2a2, length 6 → score 6−6−0=0. But deleting the two bs leaves aaaa, encoded as a4 (length 2), with deletion cost 2×1=2 → score 6−2−2=2.

Sample Input 1

```
aaaaaaaaaa
3
```

Sample Output 1

```
7
```

Explanation 1

N=10, D=3. The whole string is one character; it encodes directly to a10 (length 3) with 0 deletions → score 10−3−0=7.